02. Problem Statement: Core Challenges in AI Agent Systems

Core Challenge: Current autonomous AI agent platforms struggle with monolithic bottlenecks, sequential execution delays, proprietary tool silos, and ungrounded execution risks when handling complex workflows.

1. Monolithic LLM Bottleneck & High Cost

- **Single-Model Overload:** One massive LLM attempts planning, tool use, and synthesis in a bloated prompt context.
- **Cost & Latency Bloat:** Heavy frontier models incur 3× higher token expense and slow turnaround.
- **What We Solve:** Route sub-tasks dynamically to cost/latency-optimal models (Groq, OpenAI, NIM) with specialist agent roles.

2. Sequential & Rigid Task Execution

- **No Parallel Concurrency:** Existing chains execute steps sequentially; independent tasks wait needlessly in line.
- **Brittle Pipeline Fragility:** A single sub-task failure crashes the entire linear pipeline with no recovery.
- **What We Solve:** Convert user goals into a dynamic DAG to execute independent tasks in parallel ($K \leq 5$).

3. Fragmented Tool Silos & Black-Box Runs

- **Proprietary Integrations:** Ad-hoc APIs make connecting daily tools (Gmail, Drive, Terminal, Local Files) difficult.
- **No Real-Time Feedback:** Users are left blind during long runs without streaming status or progress logs.
- **What We Solve:** Standardize tool connectivity via Model Context Protocol (MCP) with live Server-Sent Events (SSE).

4. Safety Hazards & Cloud Data Lock-in

- **Unguarded Actions:** Agents execute irreversible actions (sending emails, deleting files) with zero human approval.
- **Privacy & Latency Risks:** Session state stored in remote cloud databases exposes sensitive personal data.
- **What We Solve:** Human-in-the-loop approval, ORPO alignment, and zero-latency local browser database (`PrismDB`).